

Understanding the Contributions of the Milky Way and M31 to the Merger Remnant Shape (Research Assignment 7)

CATHERINE SNYDER ^{1,*}

SUBMITTED MAY 8TH, 2025

¹ *University of Arizona*

ABSTRACT

This paper looks at how the contributions of two merging galaxies interact and mix during a merger. This is important to galaxy evolution because it helps to understand how a remnant finds a new equilibrium after a merger, which helps to understand how galaxies form and evolve. The paper utilizes the N-body simulation of the merger LMC and SMC employed by van Der Marel et al to explore this topic. This paper specifically looks at the contributions and intermixing of the Milky Way and M31 particles to the final remnant of their merger. The work done found that although some intermixing of particles occurs, past a certain radius the particles tend to stay in distinct areas corresponding to the original galaxies. The radius at which the remnant was well mixed, at the end of the simulation, was 588 kpc. The findings in this paper help us to understand how particles from galaxies interact, and could lead to insights about how new system equilibriums are established after mergers.

Keywords: Local Group (1) — Dark Matter Halo (1) — Halo Shape (1) — Halo Spin (1) — Galaxy (1) — Relaxation Time (1) —

1. INTRODUCTION

The area of galaxy evolution this research will investigate is the impact that galaxy mergers have on dark matter halos. This research will specifically look at mergers in the **Local Group** which is the gravitationally bound systems of galaxies of which the Milky Way is a member. Dark matter is believed to be dispersed throughout the universe in an almost web-like structure with a vast system of filaments. **Dark Matter Halos** are the areas in the galaxy where there is a higher density of dark matter compared to the average density of dark matter in the universe. These dark matter halos are virialized dark matter distributions that are differentiated from the expansion of the universe [B. Willman & J. Strader \(2012\)](#). The overall shape that these halos take is the **Halo Shape**, and the rotation of the halo is the **Halo Spin**. These halos are theorized to occur when multiple filaments intersect with one another. These dark matter halos are significant because they

gravitationally attract baryons and are where galaxies are able to form [C. Frenk & S. White \(2012\)](#). When two galaxies merge, there is a time it takes the system to return to equilibrium after the merger takes place which is the **Relaxation Time** [J. F. Navarro et al. \(2004\)](#).

A **Galaxy** is a system of stars that are gravitationally bound and have physical properties that cannot be explained by just the stars, baryons, and Newton's laws of gravity alone [B. Willman & J. Strader \(2012\)](#). We define galaxies as being made of stars and dark matter. Because these dark matter halos are thought of as the basic unit of cosmological structure, understanding dark matter halos is intrinsic to understanding cosmology. Researching the structure of dark matter halos, as well as halo formation and halo growth can give key insights into **Galaxy Evolution**, which is the formation and life cycle of a galaxy. Understanding halo growth requires understanding how these structures are impacted when galaxies merge. Understanding how the contributions of the two galaxies interact, and understanding the relaxation time of a galaxy can help to understand past, present, and future galaxy mergers.

The main understanding of dark matter halo mergers comes from cosmological simulations of mergers in near perfect settings. So far, the general understanding finds that the size and spin of the halo remnant are dependent on the energy of the orbit and the angular momentum of the orbit respectively. The shape of the remnant is primarily dependent on the initial separation between the two halos as well as the spin of the mergers [N. E. Drakos et al. \(2019a\)](#). as seen in Figure 1. In Figure 1 we also get an idea of how the contributions from each galaxy spread depending on if the orbits are radial or tangential. Although the relationship between energy of a system and the spread (how compacted/extended) of a remnant appears to be direct, the correlation between the energy of a system and the radius of the scaled density profile of the remnant appears to be an inverse relationship [N. E. Drakos et al. \(2019b\)](#). The radius of convergence for galaxies is defined as being the radius at which the relaxation time is equal to the age of the universe [J. F. Navarro et al. \(2004\)](#).

A lot of the understanding of dark matter halo mergers comes from simulations done with equal mass binary systems. This is a simplified setting for mergers, as it does not take into account mergers between different sized halos or mergers between more than two halos. These simulations do not take into account halo substructures [N. E. Drakos et al. \(2019a\)](#). One of the largest issues still being examined from these simulations is what causes the central density of merger remnants to decrease, because the current simulations see an increase in central density that does not match the drop seen in central density of halos as they grow [N. E. Drakos et al. \(2019b\)](#). Other questions left open include how the relationships and general findings of these merger simulations might change when including more complicated and more common types of mergers, such as mergers between more than two halos.

2. PROJECT

In this paper, we will examine how the contributions from the two original galaxies disperse during and after the merger. We will specifically look into the intermixing of the particles from the two different galaxies.

This project will address the question of if mergers in complicated systems with more than one galaxy influencing the merger, will coincide with the findings from the simulations of simplified mergers. It will also look into how the particles from a specific galaxy diffuse through the remnant after the merger.

Understanding what factors influence the contributions of the original galaxies as well as how they intermix can give us insight into relaxation rates of mergers

as well as how newly formed galaxies reach equilibrium after a merger. It is important to galaxy evolution because it helps us understand how the galaxies in our universe evolve over time. Understanding merger remnants can help us to understand the future of younger galaxies, as well as give us insights into older galaxies. Overall, understanding the shape of the merger gives us insight into the life cycles of galaxies.

3. METHODOLOGY

The simulations used in this project are from the work done on interactions between the Milky Way galaxy and the LMC(Large Magellanic Cloud) and SMC(Small Magellanic Cloud). [R. P. van der Marel et al. \(2012\)](#) These simulations are done using an N-body simulation which is a simulation of a system of particles and the forces acting on and between the particles in that system. The N-body system used in van Der Marel et al employs smoothed-particle hydrodynamic code, and is modeled with three different types of particles from the stellar bulge, stellar disk, and dark matter halo. This simulation runs approximately 12 Gyr.

The particle types this research will be using for analysis are the Type 1 dark matter or halo particles. This research will be looking at lower resolution(VLowRes) snapshots during the merger of the two galaxies, so in roughly 4.5 Gyr, which correlates to snapshots 235 and beyond. This research will compare multiple snapshots during the merger, including snapshots 235, 335, 435, 535, 635, and 735 to compare how the dispersion of the particles from the two original galaxies, changes over time.

The research questions will be addressed by creating a histogram that shows the particle contributions from each of the original galaxies over time. The way this is done is by making another histogram of the remnant that shows the ratio of the Milky Way divided by the total number of particles of the system, and the ratio of the M31 particles divided by the total system particles. The histogram of this ratio illustrates the areas dominated by Milky Way particles and the areas dominated by M31 particles in the remnant. Generating this histogram for the different snapshots shows us how the contributions from each galaxy get distributed over time after the merger. We will define any mixing ratio between 0.4 and 0.6 to be "well-mixed". The most balanced ratio of M31 particles to Milky Way particles will be 0.5. To get the mixing number we use the equation: $2 * (0.5 - |r - 0.5|)$. The advantage of using the mixing number is that it is near 1 when the particles are "well-mixed" and near 0 when they are composed completely of particles from one galaxy. Where r is the

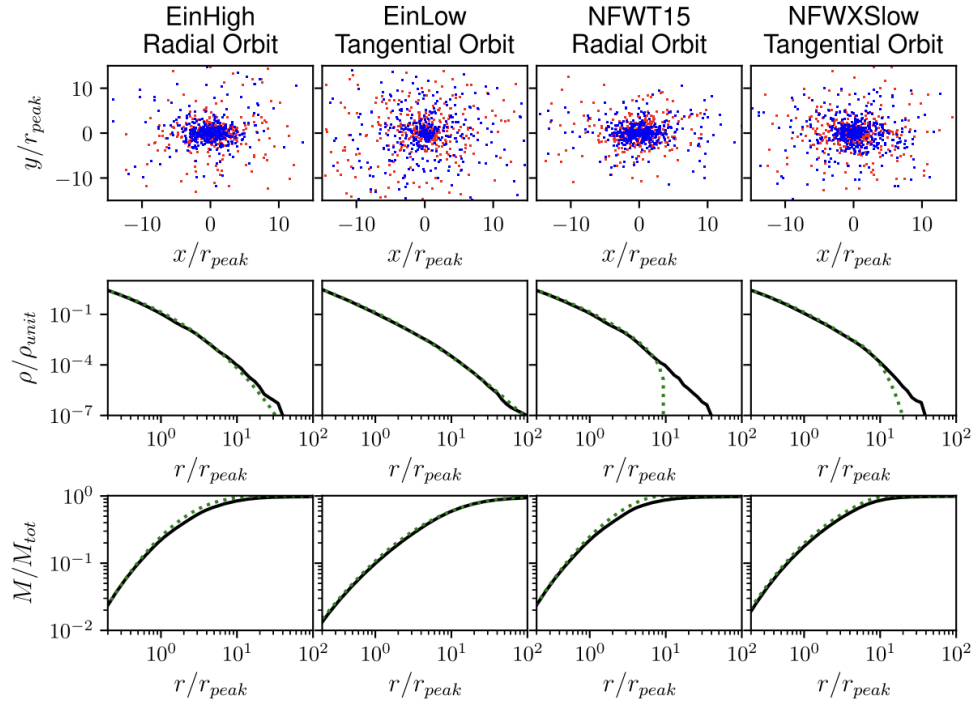


Figure 1. Figure 10 from Drakos + MNRAS, 2019a, 487, 993. This figure shows the results from the paper’s merger simulations. The top graphs are the particles of the remnants, the middle graphs are the density profiles, and the bottom graphs are the mass profiles. These results show the relations between radial orbits, tangential orbits, and where the contributions of each galaxy end up in after a merger. The red and blue particles represent the two different galaxies. This figure shows that there will be intermixing between the two galaxies but will have mainly distinct areas.

ratio of original galaxy particles to total system particles. The second figure will be a similar histogram to the first figure, with a look at the mixing number of the particles in the remnant. The third figure will look at the total number of “well mixed” particles divided by the total number of particles at that radius (referred to as the mixing number ratio on the graph), compared to the radius or the remnant. With this third graph, we can determine a “well-mixed” radius where the mixing number ratio is above 0.25. These three figures are rotated so the z-axis is aligned with the angular momentum vector of the galaxy. The models below represent the questions being investigated by this project:

Due to the path of collision the Milky Way and M31 are on, the expected outcome of this research is to find a 3-dimensional simulation with the qualities similar to the radial orbit remnants shown in Figure 10 from Drakos et al. The anticipated end result will have particles from each of the galaxies intermixing, but with one galaxy’s particle type dominating the center of the remnant while the other galaxy’s particle types become more prominent the further you get from the center of the remnant.

4. RESULTS

Figure 3 shows the histogram of the ratio of Milky Way to M31 particles. This illustrates what areas of the remnant are dominated by the Milky Way particles and what areas are dominated by the M31 particles. Figure 3 shows these ratio histograms throughout different snapshots. This shows how the contributions from the two original galaxies changes during the merger. The figure shows histograms from snapshots: 235, 335, 435, 535, 635, 735. Figure 3 displays that although the particles from the Milky Way and M31 combine during the remnant, overall the contributions from each original galaxy remain distinct and do not intermix completely.

Figure 4 shows a histogram mixing number (defined in the Methodology section) of the particles in the remnant. This figure is similar to Figure 3 but with the gradient going from red which is the most well-mixed particles, to purple which is the least well-mixed particles. The figure shows histograms from snapshots: 235, 335, 435, 535, 635, 735. The cut off of well-mixed is established at a mixing number greater than 0.8. With this graph we were able to see how well mixed the particles were throughout the remnant.

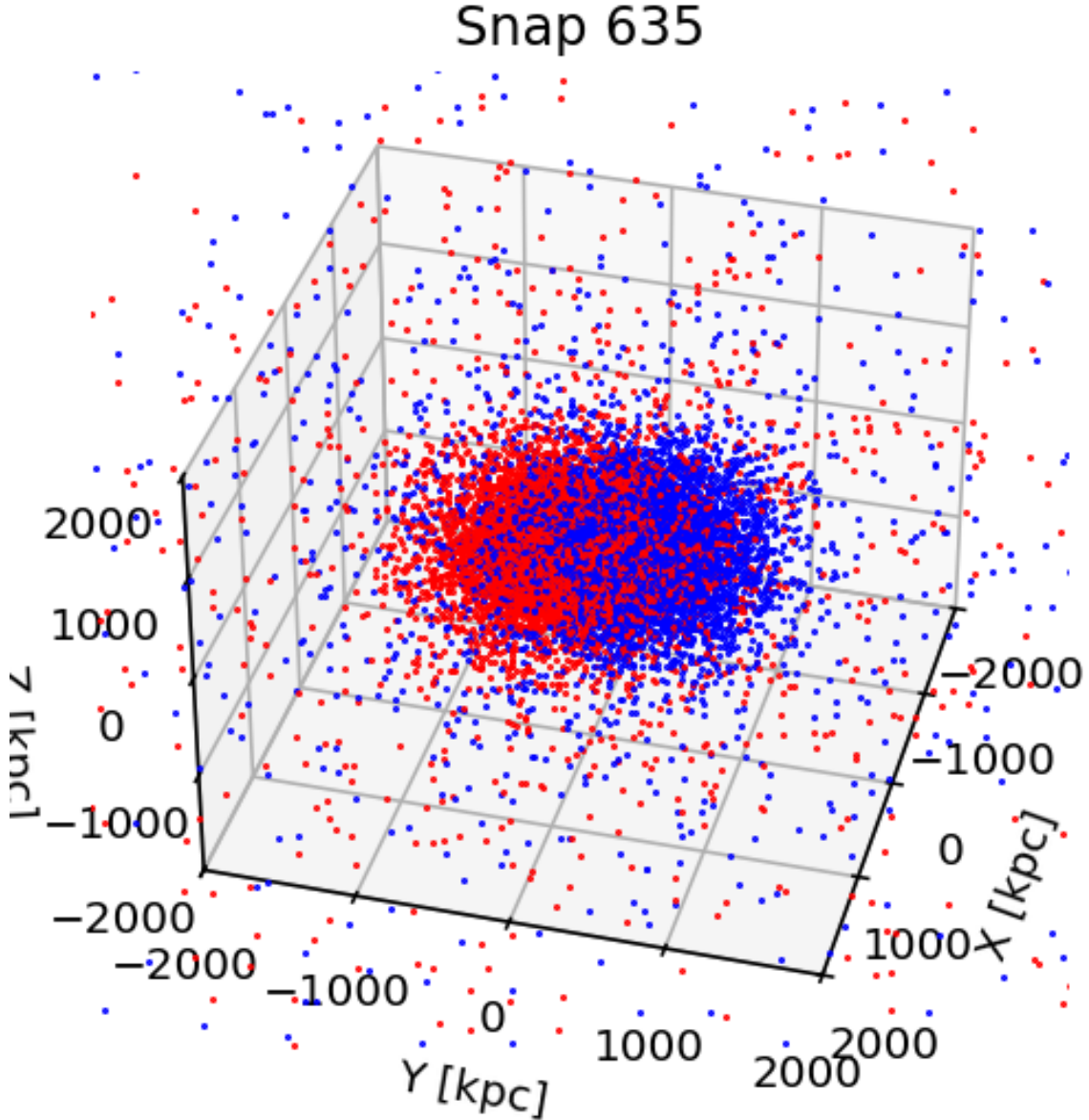


Figure 2. The figure is a three dimensional plot of a sample of the particles before during the merger. The red particles represent the M31 galaxy and the blue particles represent the Milky Way galaxy. These illustrate the investigation this project is doing into the contributions from the two original galaxies to the final remnant.

Figure 5 shows the normalized mixing ratio (defined in the Methodology section), compared to the radius of the remnant. This shows the relative fraction of volume that has "well-mixed" particles at that radius. Comparing Figure 5 to Figure 4 we determined a "well-mixed" radius is where the mixing number ratio is above 0.25. The bar graph has a line showing the "well-mixed radius" at the radius where the ratio is greater than 0.25. This shows how the "well-mixed" radius changes during the merger. The figure shows bar graphs from snapshots: 235, 335, 435, 535, 635, 735. Figure 5 shows

towards the end of the simulation, at snapshot 735, the "well-mixed ratio" is 588 kpc.

5. DISCUSSION

Figure 3 shows that although there is some intermixing of particles from the two original galaxies during the merger, they tend to stay grouped mainly in sections corresponding to their original galaxy. The two sections with particles dominating from Milky Way and M31 stayed on separate sides of the remnant, and the center of the remnant was where the most equal intermixing occurred. This disproved the hypothesis that one galaxy's particles would dominate in the center.

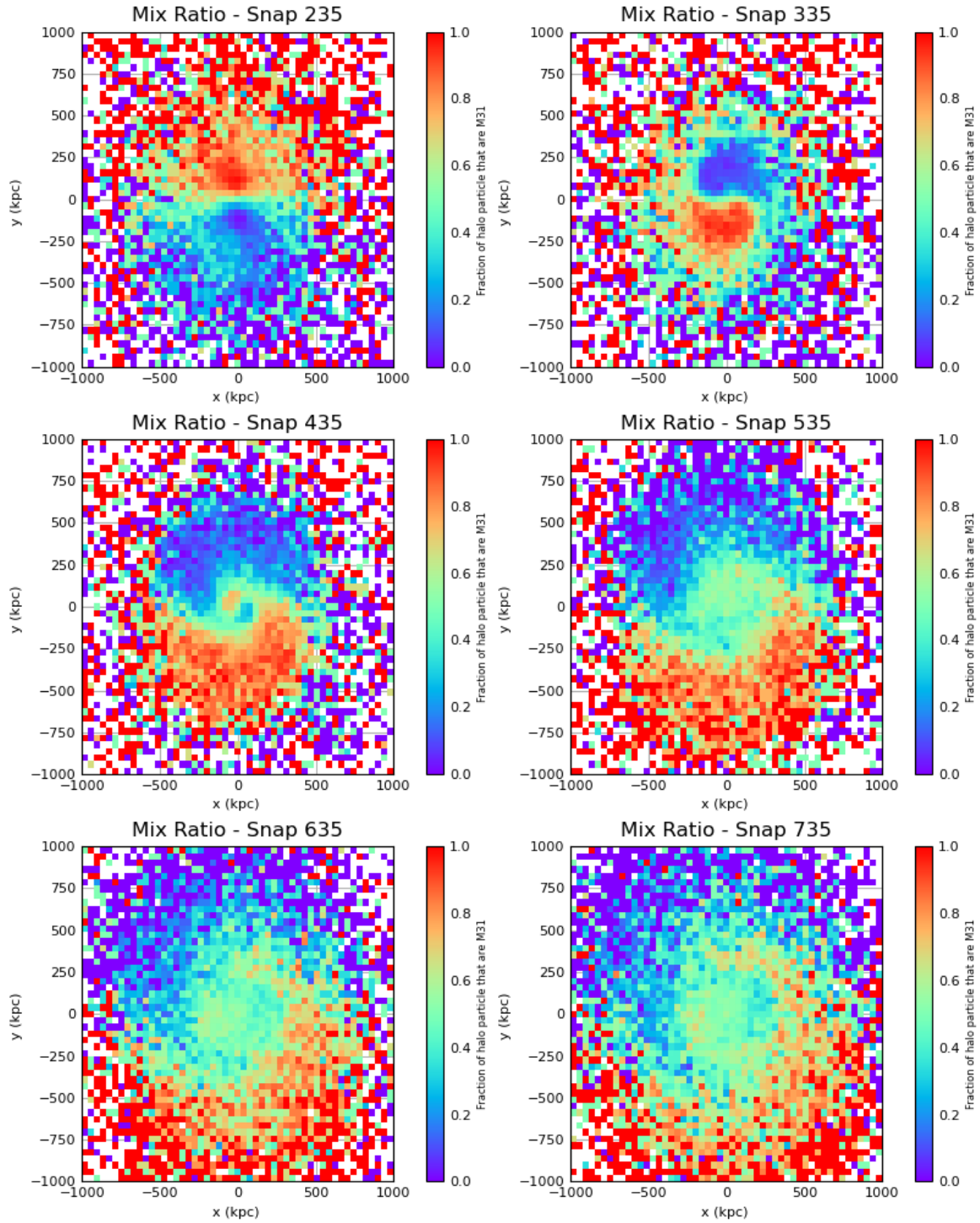


Figure 3. The figure shows the particle distribution over time. The color gradient goes from the red end which is M31 dominated, to the blue end which Milky Way dominated. This figure shows that full intermixing does not occur and comparable sections of the remnant remain dominated by each of the original galaxies' particles.

The results from Figure 5 show where the radius of intermixing is throughout the merger. Towards the end of the simulation, we find the remnant to have a "well-mixed radius" of around 600 kpc. The radius can give

insights into how the system relaxes which would help understand the relaxation rate of the galaxy.

The uncertainties from Figures 3-5 analysis come from the fact that the analysis uses low resolution snapshots. Using higher resolution files would make the histograms

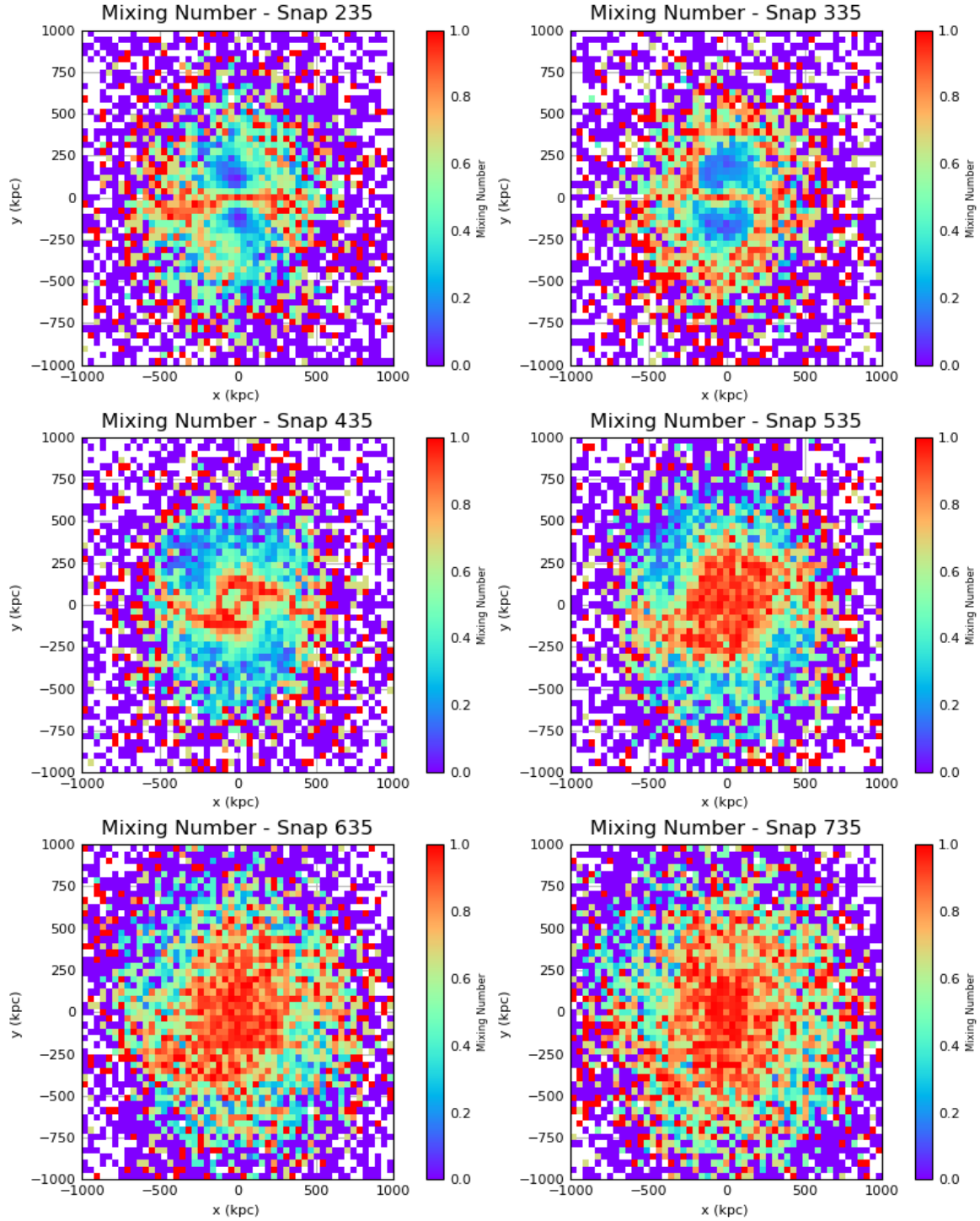


Figure 4. The figure the mixing number of the remnant. This figure shows how much intermixing occurs and gives us an idea of where well-mixed particles tend to be in the remnant.

more accurate, as we would have more precise particle data. Another uncertainty in this analysis comes from the fact the particles and influence of the M33 galaxy were excluded from the results. The halo particles from the M33 may have an impact on the mixing of these galaxies that cannot be seen in the current figures. Fur-

thermore, the code used to make these figures requires that there is an equal number of particles from each of the galaxies. If one galaxy has more particles, it would skew the ratios and impact the final results. Therefore the analysis methods for analyzing particle contributions

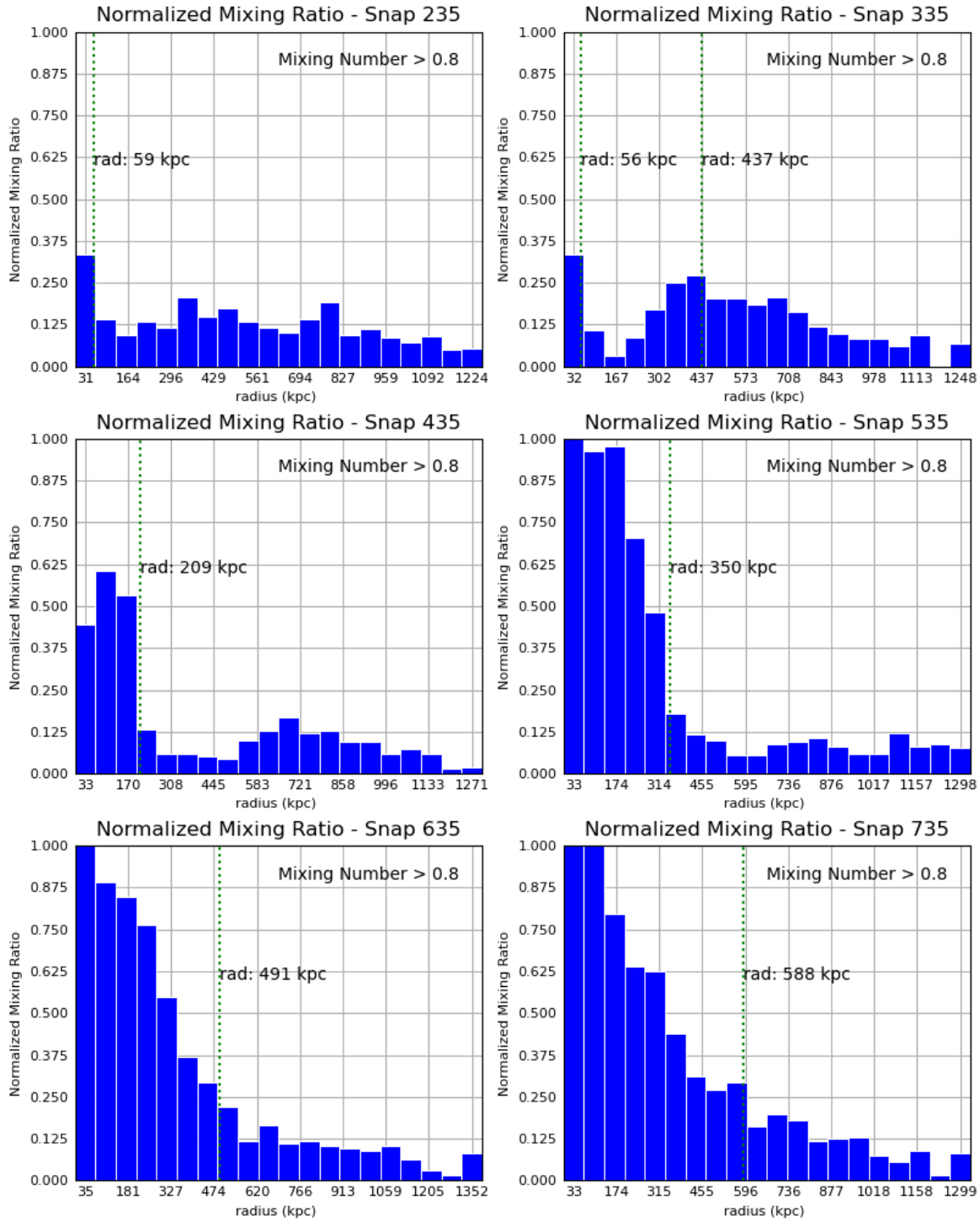


Figure 5. Figure 5 shows mixing number (defined in the Methodology section), compared to the radius or the remnant. The bar graph has a line showing the "well-mixed radius" at the radius where the ratio is greater than 0.25. This shows how the "well-mixed" radius changes during the merger. The figure shows bar graphs from snapshots: 235, 335, 435, 535, 635, 735. Figure 5 shows towards the end of the simulation, at snapshot 735, the "well-mixed ratio" is 588 kpc.

would have to be altered to look at mergers of galaxies with larger differences in size.

6. CONCLUSIONS

This paper looks at how the contributions of two merging galaxies interact and mix during a merger. This is important to galaxy evolution because it helps to un-

derstand how a remnant finds a new equilibrium after a merger, which helps to understand how galaxies form and evolve. The paper utilizes the N-body simulation of the merger LMC and SMC employed by van Der Marel et al to explore this topic. This paper specifically looks at the contributions and intermixing of the Milky Way and M31 particles to the final remnant of their merger.

One of the key findings in this analysis is that the contributions of the two galaxies do intermix at the center of the galaxy, but after a certain radius the intermixing is less prominent and distinct areas of particles correspond to the original galaxies the particles came from. This finding helps us understand how the particles interact with one another as they settle to an equilibrium. This finding did not agree with the hypothesis because one original galaxy's particles did not make up the center of the remnant.

Another finding of this paper was the "well-mixed radius" of 588 kpc that was found in snapshot 735. This radius gives us an understanding of where the particles of the two original galaxies intermix. The change in this radius over times gives insights into how the remnant relaxes over time.

Future direction this work could go is including the M33 halo particles in the system to see how they inter-

act with the other remnant contributions. Other applications could look into finding the "well-mixed radius" of merger with more galaxies or with galaxies of different sizes. Other work could be investigate a connection between the relaxation time of the system and the "well-mixed radius" of the system.

7. ACKNOWLEDGEMENTS

This work has been supported by Dr. Gurtina Besla in her lectures, labs, and assignments. Her support with technical issues and help narrowing down focus of analysis are also deeply appreciated. The technical support and feedback of Himansh Rathore is gratefully acknowledged. Support with troubleshooting, code support, and feedback from Michael Snyder is also gratefully acknowledged. This work made use of the following software packages: `astropy` (Astropy Collaboration et al. 2013, 2018, 2022), `Jupyter` (F. Perez & B. E. Granger 2007; T. Kluyver et al. 2016), `matplotlib` (J. D. Hunter 2007), `numpy` (C. R. Harris et al. 2020), `python` (G. Van Rossum & F. L. Drake 2009), and `scipy` (P. Virtanen et al. 2020; R. Gommers et al. 2025).

Software citation information aggregated using [The Software Citation Station](#) (T. Wagg & F. S. Broekgaarden 2024; T. Wagg et al. 2024).

REFERENCES

- Astropy Collaboration, Robitaille, T. P., Tollerud, E. J., et al. 2013, *A&A*, 558, A33, doi: [10.1051/0004-6361/201322068](https://doi.org/10.1051/0004-6361/201322068)
- Astropy Collaboration, Price-Whelan, A. M., Sipőcz, B. M., et al. 2018, *AJ*, 156, 123, doi: [10.3847/1538-3881/aabc4f](https://doi.org/10.3847/1538-3881/aabc4f)
- Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, *ApJ*, 935, 167, doi: [10.3847/1538-4357/ac7c74](https://doi.org/10.3847/1538-4357/ac7c74)
- Drakos, N. E., Taylor, J. E., Berrouet, A., Robotham, A. S. G., & Power, C. 2019a, *MNRAS*, 487, 993, doi: [10.1093/mnras/stz1306](https://doi.org/10.1093/mnras/stz1306)
- Drakos, N. E., Taylor, J. E., Berrouet, A., Robotham, A. S. G., & Power, C. 2019b, *MNRAS*, 487, 1008, doi: [10.1093/mnras/stz1307](https://doi.org/10.1093/mnras/stz1307)
- Frenk, C., & White, S. 2012, *Annalen der Physik*, 524, 507–534, doi: [10.1002/andp.201200212](https://doi.org/10.1002/andp.201200212)
- Gommers, R., Virtanen, P., Haberland, M., et al. 2025,, v1.15.3 Zenodo, doi: [10.5281/zenodo.15366870](https://doi.org/10.5281/zenodo.15366870)
- Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, *Nature*, 585, 357, doi: [10.1038/s41586-020-2649-2](https://doi.org/10.1038/s41586-020-2649-2)
- Hunter, J. D. 2007, *Computing in Science & Engineering*, 9, 90, doi: [10.1109/MCSE.2007.55](https://doi.org/10.1109/MCSE.2007.55)
- Kluyver, T., Ragan-Kelley, B., Pérez, F., et al. 2016, in *Positioning and Power in Academic Publishing: Players, Agents and Agendas*, ed. F. Loizides & B. Schmidt, IOS Press, 87 – 90
- Navarro, J. F., Hayashi, E., Power, C., et al. 2004, *MNRAS*, 349, 1039, doi: [10.1111/j.1365-2966.2004.07586.x](https://doi.org/10.1111/j.1365-2966.2004.07586.x)
- Perez, F., & Granger, B. E. 2007, *Computing in Science and Engineering*, 9, 21, doi: [10.1109/MCSE.2007.53](https://doi.org/10.1109/MCSE.2007.53)
- van der Marel, R. P., Besla, G., Cox, T. J., Sohn, S. T., & Anderson, J. 2012, *The Astrophysical Journal*, 753, 9, doi: [10.1088/0004-637X/753/1/9](https://doi.org/10.1088/0004-637X/753/1/9)
- Van Rossum, G., & Drake, F. L. 2009, *Python 3 Reference Manual* (Scotts Valley, CA: CreateSpace)
- Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, *Nature Methods*, 17, 261, doi: [10.1038/s41592-019-0686-2](https://doi.org/10.1038/s41592-019-0686-2)
- Wagg, T., Broekgaarden, F., & Gültekin, K. 2024,, v1.2 Zenodo, doi: [10.5281/zenodo.13225824](https://doi.org/10.5281/zenodo.13225824)
- Wagg, T., & Broekgaarden, F. S. 2024, arXiv e-prints, arXiv:2406.04405. <https://arxiv.org/abs/2406.04405>
- Willman, B., & Strader, J. 2012, *The Astronomical Journal*, 144, 76, doi: [10.1088/0004-6256/144/3/76](https://doi.org/10.1088/0004-6256/144/3/76)